

Question Bank (ADA)

UNIT – 1

1. What are the features of R programming language and what are application? (what is R and why learn R)
2. What are the important packages in R. also mention the common R function for dealing with dataset. (Table 2 package and use, function and purpose).
3. What are the basic data types and data structure in R.(explain)
4. Explain with example different date formatting in R.(page 29 practice 4)
5. Explain with example factors in R. (page 24 practice 2)
6. Explain with example. how character manipulation R. (page 28 practice 3)
7. Demonstrate reshaping of data in R using pivot longer and pivot wider function. (page 26-27 Question-2).
8. what are the different ways of doing data aggregation in R? (only code - way 1,2,3,4)
9. what are the different function of reading and writing function in R. (page no.7-8 long description)
10. what are the different function use to handle databases in R. (page-11 write in long description)

UNIT – 2

1. What are the different types of discrete distribution and what are there function in R. (page 30 – binomial ,Poisson)
2. What are the different function in R for continuous probability distribution. (page-30,31[dnorm,pnorm,rnorm,uniform,exponential])
3. What are the non-visualization based techniques or functions in R for Descriptive statistics. (page – 6, only description)
4. What are the visualization based techniques or function in R for analyzing descriptive statistics.(page-33 code, barplot,lineplot,scatterplot with description).
5. What are the different types of t-test. give example with R code.(page-34, 4 types)
6. Give the comparison of different type of hypothesis testing.(page-35)
7. When should we make use of which type of hypothesis testing.explain?
8. Write short note on krushkal-wallis test and explain its working with R code.(page-37)
9. What are the different types of diagnostic plots to analyze multiple linear regression model.(page-15) (code-40) [build 2 line code ,setup 2*2]
10. Write short note on survival analysis. (page -17,18)

11. Write short note on poisson regression. explain poisson regression model.(only theory)
12. Write short note on power and computation of sample size. (page-13-14, only theory)